House Price Prediction Using Machine Learning Regression Models: A Comparative Study

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ABSTRACT

Accurate prediction of residential property values is a critical challenge in real estate analytics, urban planning, and financial risk assessment. This paper presents a comparative study of six machine learning regression models — Linear Regression, Ridge Regression, Lasso Regression, Decision Tree, Random Forest, and Support Vector Regression (SVR) — applied to the widely-used Boston Housing Dataset. Each model is evaluated using standard metrics including the Coefficient of Determination (R^2), Mean Absolute Error (MAE), and Mean Squared Error (MSE). The Random Forest Regressor achieved the highest predictive accuracy with an R^2 score of approximately 0.87, outperforming all other models. The best-performing model is deployed as a Flask-based web application that accepts user-provided property features and returns an instant house price prediction. Results demonstrate that ensemble methods significantly outperform linear and kernel-based approaches on this dataset.

Keywords — House Price Prediction, Boston Housing Dataset, Machine Learning, Random Forest, Flask, Regression Analysis, Feature Engineering

I. INTRODUCTION

The prediction of real estate prices is a problem of substantial economic importance. Buyers, sellers, investors, and policymakers all depend on accurate property valuations to make informed decisions. Traditionally, property appraisal has relied on manual assessment by expert valuators — a process that is both time-consuming and prone to subjective bias.

The emergence of machine learning (ML) has opened new avenues for automated, data-driven price prediction. By learning complex nonlinear relationships between property attributes and market prices, ML models can achieve prediction accuracy that rivals or surpasses traditional methods. The Boston Housing Dataset, introduced by Harrison and Rubinfeld [1], remains a canonical benchmark for regression tasks in the ML community, providing 506 instances across 13 socioeconomic and structural features.

This study makes three primary contributions. First, we evaluate and compare six regression algorithms — Linear Regression, Ridge, Lasso, Decision Tree, Random Forest, and SVR — under a consistent experimental protocol. Second, we identify Random Forest as the best-performing model through quantitative comparison. Third, we demonstrate end-to-end deployment of the trained model as a production-ready Flask web application, making the prediction system accessible to non-technical users.

II. RELATED WORK

House price prediction has been studied extensively using both classical statistical methods and modern machine learning approaches. Limsombunchai [2] applied hedonic pricing models using artificial neural networks (ANNs) and demonstrated that nonlinear models outperform traditional linear regression. Gu et al. [3] compared multiple regression methods on the Boston dataset and found that ensemble methods consistently achieve lower prediction error.

Recent work has explored gradient boosting approaches such as XGBoost [4] and LightGBM for house price prediction on large real-world datasets. However, for the standard Boston benchmark, Random Forest remains highly competitive and provides good model interpretability through feature importance scores. Our study builds upon this body of work by providing a direct, reproducible comparison of six models with a web deployment pipeline.

III. DATASET DESCRIPTION

The Boston Housing Dataset was originally collected by Harrison and Rubinfeld in 1978 as part of a study on the relationship between clean air and house prices. It contains 506 samples describing suburbs of Boston, Massachusetts, with 13 input features and one continuous target variable (MEDV — Median value of owner-occupied homes in thousands of US dollars).

Feature	Description	Type
CRIM	Per capita crime rate by town	Float
ZN	% residential land zoned >25,000 sq ft	Float
INDUS	% non-retail business acres per town	Float
CHAS	Charles River dummy variable (1=bounds river)	Binary
NOX	Nitric oxide concentration (ppm×10)	Float
RM	Average number of rooms per dwelling	Float
AGE	% owner-occupied units built before 1940	Float
DIS	Weighted distance to employment centres	Float
RAD	Index of highway accessibility	Int
TAX	Property-tax rate per $10,000	Float
PTRATIO	Pupil-teacher ratio by town	Float
B	$1000(Bk-0.63)^2$ — Black population index	Float
LSTAT	% lower status population	Float
MEDV	Median home value in $1,000s (Target)	Float

Table I: Boston Housing Dataset Features

The dataset contains missing values in six columns: CRIM, ZN, INDUS, CHAS, AGE, and LSTAT (20 missing values each). Missing values were imputed using the column mean for CRIM, ZN, INDUS, and CHAS; the standard deviation for AGE; and the median for LSTAT. These strategies reflect the statistical distribution characteristics of each feature.

IV. METHODOLOGY

A. Exploratory Data Analysis

Prior to model training, exploratory data analysis (EDA) was conducted to understand the relationships between features and the target variable. Scatter plots revealed strong negative correlations between MEDV and both LSTAT (lower status population %) and CRIM (crime rate). Conversely, RM (average rooms) showed a strong positive correlation with MEDV. A correlation heatmap confirmed that LSTAT ($r = -0.74$) and RM ($r = +0.70$) are the most informative predictors.

B. Data Preprocessing

The preprocessing pipeline consisted of three steps:

- Missing value imputation as described in Section III.
- Feature-target split: features X = all columns except MEDV; target y = MEDV.
- 80/20 train-test split with random_state=400 for reproducibility.
- Feature standardization using StandardScaler fitted exclusively on the training set to prevent data leakage.

C. Models

Six regression models were trained and evaluated. Linear Regression provides an interpretable baseline. Ridge and Lasso add L2 and L1 regularization respectively ($\alpha = 0.01$) to penalize large coefficients and reduce overfitting. The Decision Tree Regressor partitions the feature space recursively but is prone to overfitting on training data. The Random Forest Regressor is an ensemble of decision trees that aggregates predictions to reduce variance. Support Vector Regression (SVR) maps inputs to a high-dimensional feature space using a kernel function to find an optimal regression hyperplane.

D. Evaluation Metrics

Models were compared using three standard regression metrics:

- R^2 (Coefficient of Determination): proportion of variance explained; higher is better (max 1.0).
- MAE (Mean Absolute Error): average absolute prediction error in $1,000 units; lower is better.
- MSE (Mean Squared Error): penalizes large errors more than MAE; lower is better.

V. EXPERIMENTAL RESULTS

All six models were trained on 80% of the data (404 samples) and evaluated on the remaining 20% (102 samples). Table II summarizes the performance metrics for each model.

Model	Train Score (%)	Test Score (%)	R^2	MAE	MSE
Random Forest ⭑	97.1	87.2	0.872	2.31	10.84
Linear Regression	73.4	74.1	0.741	3.48	22.10
Ridge (α=0.01)	73.4	74.1	0.741	3.47	22.08
Lasso (α=0.01)	73.3	74.0	0.740	3.50	22.20
Decision Tree	100.0	69.3	0.693	3.22	28.90
SVR	79.5	63.2	0.632	3.96	33.20

Table II: Model Performance Comparison on Test Set

The Random Forest Regressor achieved the highest R^2 of 0.872, indicating it explains approximately 87% of the variance in house prices. Its MAE of 2.31 ($2,310) represents a significant improvement over the linear baselines. The Decision Tree achieved perfect training accuracy (100%) but lower test performance (69.3%), a clear indication of overfitting. SVR performed the weakest, likely due to its sensitivity to feature scale and the absence of hyperparameter tuning in this study.

The three linear models (Linear Regression, Ridge, Lasso) achieved nearly identical performance (~74%), confirming that the Boston Housing data contains nonlinear relationships that linear models cannot fully capture. The minor difference between Ridge and Lasso reflects the small regularization strength ($\alpha = 0.01$) used.

VI. WEB APPLICATION DEPLOYMENT

The trained Random Forest model was serialized using Python's pickle library and deployed as a Flask web application. The application exposes a form-based interface (index.html) where users input four key property features (CRIM, ZN, INDUS, CHAS) and receive an instant MEDV prediction. The remaining nine features are automatically populated with dataset-derived mean values, ensuring the model always receives a complete 13-dimensional input vector.

The application architecture follows an MVC pattern:

- Model Layer: Pickle-serialized RandomForestRegressor loaded at startup.

- Controller Layer: Flask routes handle GET (render form) and POST (run prediction) requests.
- View Layer: Jinja2 HTML template renders the form and displays the predicted price inline.

Additional API endpoints (/predict/all, /metrics) allow programmatic access from external clients, enabling integration with third-party analytics pipelines.

VII. DISCUSSION

The results confirm the well-established advantage of ensemble methods over individual learners for tabular regression tasks. Random Forest's ability to average across many de-correlated trees reduces both bias and variance, yielding the best generalization. The gap between its training score (97.1%) and test score (87.2%) suggests mild overfitting, which could be addressed by tuning the number of estimators and max depth.

A notable limitation of this study is the omission of hyperparameter optimization. Grid search or Bayesian optimization over SVR's kernel and C parameters, for example, could substantially improve its test performance. Similarly, tuning Random Forest's n_estimators, max_features, and min_samples_split may further close the train-test gap.

From a feature importance perspective, LSTAT and RM together account for approximately 69% of the Random Forest's predictive weight, aligning with the EDA findings. This supports the interpretation that socioeconomic status and dwelling size are the dominant drivers of house value in the Boston dataset.

VIII. CONCLUSION

This paper presented a systematic comparison of six machine learning regression models for house price prediction using the Boston Housing Dataset. The Random Forest Regressor was identified as the superior model with an R^2 of 0.872 and MAE of 2.31, and was subsequently deployed as a Flask web application accessible to end users. Future work will explore advanced ensemble methods (XGBoost, LightGBM), hyperparameter tuning, and deep learning architectures such as multi-layer perceptrons for further accuracy gains.

REFERENCES

[1] D. Harrison and D. L. Rubinfeld, "Hedonic housing prices and the demand for clean air," Journal of Environmental Economics and Management, vol. 5, no. 1, pp. 81–102, 1978.

[2] V. Limsombunchai, "House Price Prediction: Hedonic Price Model vs. Artificial Neural Network," New Zealand Agricultural and Resource Economics Society Conference, 2004.

[3] J. Gu, Z. Zhu, and L. Jiang, "House Price Forecasting with Neural Networks," Intelligent Systems in Accounting, Finance and Management, vol. 18, no. 2-3, pp. 169–186, 2011.

[4] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, pp. 785–794, 2016.

[5] L. Breiman, "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[6] A. J. Smola and B. Schölkopf, "A tutorial on support vector regression," Statistics and Computing, vol. 14, no. 3, pp. 199–222, 2004.

[7] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.